

Concepts and Equations for Petroleum Engineers

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Reservoir Engineering

1. Darcy's Law

➤ Expression

$$q = -k \times \frac{A}{\mu} \times \frac{dp}{dx}$$

where

q – single phase flow rate, cm^3/sec

k – permeability, *Darcy*

A – flow area, cm^2

dp – pressure drop, *atm*

dx – flow length, *cm*

The units used in the law are called Darcy units. Note: $1\ md = 10^{-15}\ m^2$, $1\ Darcy = 1\ (\mu m)^2$

Rank of the formation based on permeability

$<1\ md$ - Very low

$1\ to\ 10\ md$ - Low

$10\ to\ 50\ md$ - Medium

$50\ to\ 200\ md$ - Average

$200\ to\ 500\ md$ - Good

$>500\ md$ - Excellent

➤ Darcy's law in field metric units

$$q = -8.64 \times 10^{-3} k \times \frac{A}{\mu B} \times \frac{dp}{dx}$$

Where q – volumetric flow rate at stock tank conditions, m^3/day

k – permeability, *mD*

B - formation volume factor of the fluid, m^3/m^3

A – flow area, m^2

dp – pressure drop, *atm*

dx – flow length, *m*

➤ Darcy's law in radial steady-state flow

$$q = 5.435 \times 10^{-2} \frac{kh(p_e - p_{wf})}{B\mu\{\ln(r_e / r_w) + S\}}$$

where

p_e – reservoir pressure at outer boundary radius r_e , *bars*

p_{wf} – bottom-hole pressure (BHP) at r_w , *bars*

r_e – outer boundary radius, *m*

r_w – wellbore radius, *m*

S – skin factor, dimensionless

h – formation thickness, *m*

➤ Darcy's law in radial flow pseudosteady-state flow

$$q = 5.435 \times 10^{-2} \frac{kh(p_e - p_{wf})}{B\mu\{\ln(r_e / r_w) - 0.75 + S\}}$$

2. OOIP/Reserves

➤ OOIP- Original Oil in Place

$$N = \int \frac{h\phi(1-S_w)}{B_o} dA$$

where

N – OOIP at standard conditions, sm^3

h – reservoir thickness, m

ϕ – reservoir porosity, *fraction*

S_w – water saturation, *fraction*

B_o – oil FVF, m^3/sm^3

Remark: all the quantities in the integrant will vary arealy in general,
So numerical integration is needed to calculate the OOIP

➤ Reserves

Defined as the future economically recoverable hydrocarbons (oil) from a reservoir
Ultimate recovery is given by

$$R = OOIP \times E_r$$

R – ultimate recovery

$OOIP$ – Original Oil-in-Place

E_r – recovery efficiency

$$\text{Reserves} = R - Q_p$$

Q_p – cumulative oil production

3. Skin Factor

- To represent formation damage around a wellbore due to drilling, completion and stimulation processes
- Definition

$$S = \left(\frac{k}{k_s} - 1 \right) \ln \frac{r_s}{r_w}$$

where

k - permeability of the undamaged reservoir, md

k_s - permeability of damage zone, md

r_s - radius of the damaged zone, m

r_w - wellbore radius, m

- The additional pressure drop due to skin effect, Δp_s

$$\Delta p_s = \frac{18.4qB\mu}{kh} S$$

where

q – volumetric oil flow rate at stock tank conditions, m^3/d

B – oil FVF, m^3/m^3

h – formation thickness, m

μ - oil viscosity, cp

Δp_s - pressure drop, atm

- Example

Assume a well has a radius $r_w = 10cm$ (0.328ft), and a damage radius $r_s = 3.328ft$, drilling mud invasion and partial perforation results in $k/k_s = 5$, the well is producing $50 m^3/d$ of oil with viscosity $1.3cp$ and FVF $1.21 m^3/m^3$, reservoir thickness at the well location is about $15m$, what would be the skin effect?

$$\text{Skin factor: } S = (5 - 1) \times \ln \frac{3.328}{0.328} = 9.3$$

$$\text{Pressure drop due to skin: } \Delta p_s = \frac{18.4 \times 50 \times 1.21 \times 1.3}{5 \times 15} \times 9.3 = 19.3 atm$$

- Notes

4. Diffusivity Equation, Flow Regimes

➤ Diffusivity equation in radial form

$$\frac{\partial^2 p}{\partial r^2} + \frac{1}{r} \frac{\partial p}{\partial r} = \frac{\phi \mu c_t}{k} \frac{\partial p}{\partial t}$$

Where

p – pressure

r – radius

t - time

ϕ – porosity

c_t – total compressibility of the porous media

k – permeability

This radial flow equation applies to any coherent units system.

➤ Transient flow, pressure, so as flow rate, changes with time t and radius r , i.e.,

$$\frac{\partial p}{\partial t} = f(r, t)$$

This flow state is also called infinite acting. It is assumed that the pressure response in the reservoir is not affected by the outer boundary.

➤ Semi steady state (or pseudo steady-state)

$$\frac{\partial p}{\partial t} = \text{const}$$

This condition is applicable to a reservoir which has been producing for a sufficient period of time so that the effect of the outer boundary has been felt.

If the well is producing at a constant flow rate q , then the constant can be expressed as

$$\frac{\partial p}{\partial t} = - \frac{q}{c \pi r_e^2 h \phi}$$

where r_e is the outer boundary radius and h is the reservoir thickness.

➤ Steady state

$$\frac{\partial p}{\partial t} = 0$$

This applies to a constant pressure boundary or the pressure is maintained in the reservoir due to natural flow influx or fluid injection, such as water injection.

5. Well Test Analysis

➤ Pressure drawdown or injectivity

$$p_i - p_{wf} = \frac{21.5q\mu B_o}{kh} \left\{ \log(t) + \log \frac{k}{\phi\mu c_t r_w^2} - 3.1 + 0.87S \right\}$$

$$\text{Permeability: } k = 21.5 \frac{q\mu B_o}{mh}$$

$$\text{Skin factor: } S = 1.15 \times \left\{ \frac{p_i - p_{1hr}}{m} - \log \left(\frac{k}{\phi\mu c_t r_w^2} \right) + 3.10 \right\}$$

Where

p_{wf} – flowing bottom hole pressure, *bar*

p_i – initial reservoir pressure, *bar*

p_{1hr} – pressure on straight line portion of semilog plot 1 hour after beginning a transient test

q – volumetric flow rate, m^3/d

μ – oil viscosity, *cp*

B_o – oil FVF, m^3/sm^3

t – time, *hours*

k – permeability of the reservoir, *md*

ϕ – reservoir porosity, *fraction*

h – reservoir thickness, *m*

c_t – total compressibility, bar^{-1}

S – skin factor, dimensionless

r_w – wellbore radius, *m*

m – the slope of the semilog straight line, *bar/cycle*

➤ Buildup or falloff

$$p_i - p_{ws}(\Delta t) = \frac{21.5q\mu B_o}{kh} \ln \frac{t_p + \Delta t}{\Delta t}$$

$$\text{Permeability: } k = \frac{21.5q\mu B_o}{mh}$$

$$\text{Skin factor: } S = 1.1513 \times \left\{ \frac{p_i - p_{1hr}}{m} - \log \left(\frac{k}{\phi\mu c_t r_w^2} \right) + 3.2275 + \log \left(\frac{t_p + 1}{t_p} \right) \right\}$$

where

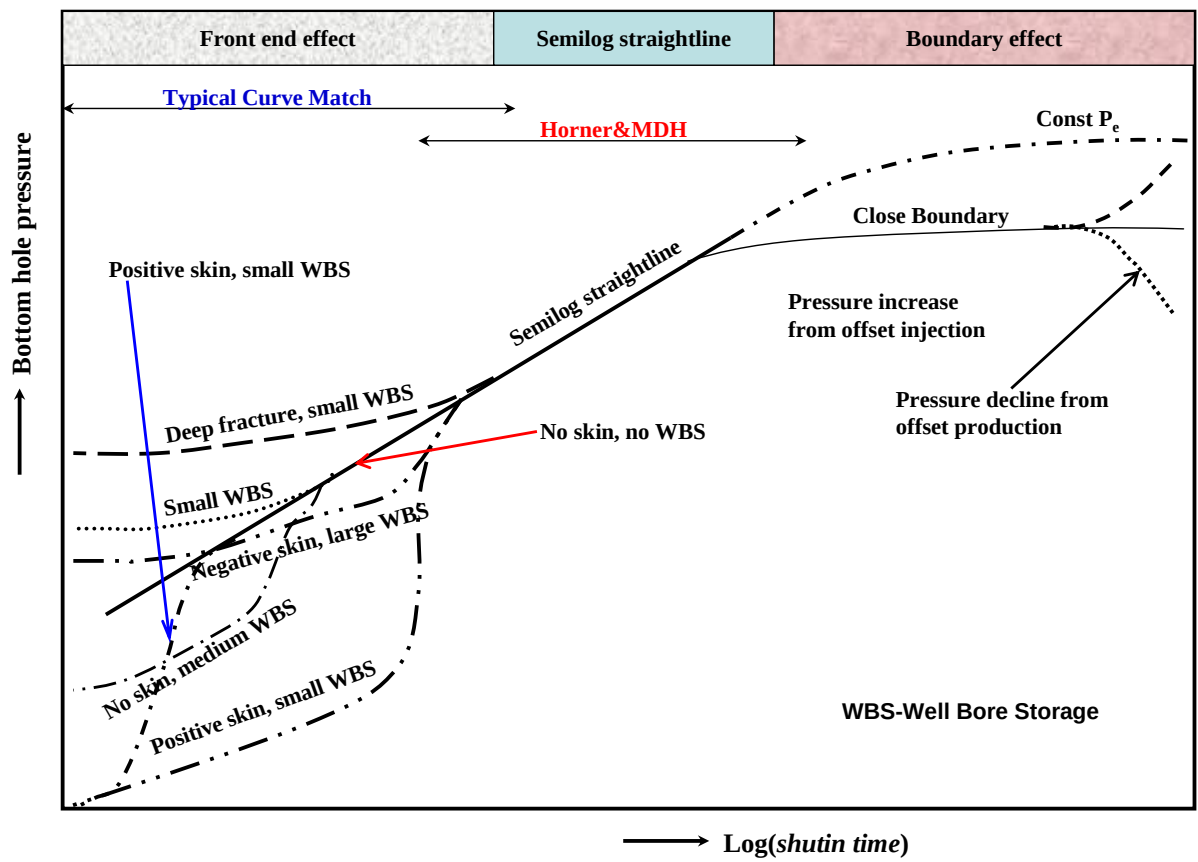
$p_{ws}(\Delta t)$ – bottom hole pressure during the buildup phase, *bar*

t_p – production or injection time, *hours*

$$t_p = \frac{\sum q}{q_{last}} \times 24 \text{ hours}$$

Δt – shut in time, *hours*

m – the slope of the straight line on a graph of p_{ws} versus $\log(t_p + \Delta t)/\Delta t$



6. Water Injection Compensation

➤ **Definition: cumulative compensation**

$$C_m (\%) = \frac{\sum q_i B_w}{\sum q_o B_o + \sum q_w B_w} \times 100\%$$

where

$\sum q_i B_w$ – cumulative water injected into the entity from the injector(s).

$\sum q_o B_o$ – cumulative oil production from the entity at reservoir conditions.

$\sum q_w B_w$ – cumulative water production from the entity at the reservoir conditions.

B_o – oil FVF.

B_w – water FVF.

➤ **Definition: instantaneous compensation**

$$c_m = \frac{q_i B_w}{q_o B_o + q_w B_w} \times 100\%$$

where

q_i – water injection rate into the entity from the injector(s).

q_o – oil production rate from the entity at surface conditions.

q_w – water production rate from the entity at the surface conditions.

B_o – oil FVF.

B_w – water FVF.

➤ **Explanation**

This parameter tells roughly how the reservoir pressure is maintained by water injection. Local practice suggests that a value of C_m between 110% to 120% would indicate good reservoir pressure maintenance by water injection.

7. Immiscible Displacement (Waterflooding)

➤ Linear water fractional flow equation, f_w

$$f_w = \frac{1 + 8.64 \times 10^{-3} \frac{kk_{ro}}{\mu_o} \frac{A}{q_t} \left[\frac{\partial P_c}{\partial L} - 0.1 \times (\rho_w - \rho_o) \sin(\alpha) \right]}{1 + \frac{\mu_w}{\mu_o} \frac{k_{ro}}{k_{rw}}}$$

or

$$f_w = \frac{1}{1 + \frac{\mu_w}{\mu_o} \frac{k_{ro}}{k_{rw}}}$$

if gravity and capillary effects can be neglected.

where

A – area of the linear system, m^2

f_w – water fractional flow, fraction

k – absolute permeability, md

k_{ro}, k_{rw} – relative permeability of oil and water, respectively

μ_o, μ_w – oil and water viscosity, cp

ρ_o, ρ_w – oil and water density, g/cm^3

L – distance along direction of flow, m

$P_c = p_o - p_w$ – capillary pressure, atm

q_t – total flow rate, $q_o + q_w$, m^3/d

α – angle of the linear system from horizon, degrees

➤ Linear frontal advance equation (Buckley-Leverett)

$$\left(\frac{\partial x}{\partial t} \right)_{S_w} = \frac{q_t}{A\phi} \left(\frac{\partial f_w}{\partial S_w} \right)_t$$

Physically, this equation says that the advance rate of a fixed water saturation, S_w , at a time, t , is equal to the total fluid velocity times the change in composition of the flowing stream. This equation can be used to derive the average water saturation of the system.

➤ Average water saturation

$$\text{At water breakthrough: } \bar{S}_{wbt} - S_{wc} = \frac{S_{wf} - S_{wc}}{f_{wf}}$$

$$\text{After water breakthrough: } \bar{S}_w - S_{w2} = \frac{1 - f_{w2}}{f'_{w2}}$$

Where S_{wc} – connate water saturation

S_{wf} – water saturation at the flood front

S_{w2} – water saturation at the producing end of the system

$f_{wf} = f_w(S_{wf})$ – water fractional flow at $S_w = S_{wf}$

$f_{w2} = f_w(S_{w2})$ – water fractional flow at $S_w = S_{w2}$

f'_{w2} – derivative of f_w at $S_w = S_{w2}$

$\bar{S}_{wbt}$ – average water saturation before breakthrough

$\bar{S}_w$ – average water saturation after water breakthrough

8. Material Balance Methods

➤ General equation

Underground = Expansion of oil
 Withdrawal + Original dissolved gas + Expansion of gas caps
 + Reduction in hydrocarbon pore volume due to connate water expansion and decrease in the pore volume.
 + Natural water influx

$$F = N(E_o + mE_g + E_{fw}) + W_e = NE_t + W_e$$

where:

F = underground withdrawal, m^3

$$= N_p [B_o + (R_p - R_s)B_g] + W_p B_w - W_i B_w - G_i B_g$$

$$= N_p [B_t + (R_p - R_{si})B_g] + W_p B_w - W_i B_w - G_i B_g$$

N_p = cumulative oil production, sm^3

B_o = oil formation volume factor, m^3/sm^3

R_s = gas in solution in oil, sm^3/sm^3

B_g = gas formation volume factor, m^3/sm^3

W_p = cumulative water production, sm^3

W_i = cumulative water injection, sm^3

G_i = cumulative gas injection, sm^3

R_p = cumulative gas oil ratio (GOR), sm^3/sm^3

N = OOIP, sm^3

$B_t = B_o + (R_{si} - R_s)B_g$

E_o = Expansion of oil and original gas in solution, m^3/sm^3

$$= B_t - B_{ti} = (B_o - B_{oi}) + (R_{si} - R_s)B_g$$

m = initial gas cap volume factor

$$= \frac{\text{initial hydrocarbon volume of the gas cap}}{\text{initial hydrocarbon volume of the oil zone}}, m^3/m^3$$

E_g = Expansion gas cap gas, m^3/sm^3

$$= B_{oi} \left(\frac{B_g}{B_{gi}} - 1 \right)$$

E_{fw} = Expansion of the connate water and reduction in the pore volume, m^3/sm^3

$$= (1 + m)B_o \left(\frac{C_w S_{wi} + C_f}{1 - S_{wi}} \right) \Delta P$$

C_w, C_f = water, formation, compressibility, respectively, atm^{-1}

S_{wi} = initial water saturation, fraction

ΔP = pressure drop, atm

W_e = cumulative natural water influx, m^3

$$= U \times S(p, t)$$

U = aquifer constant, m^3/atm

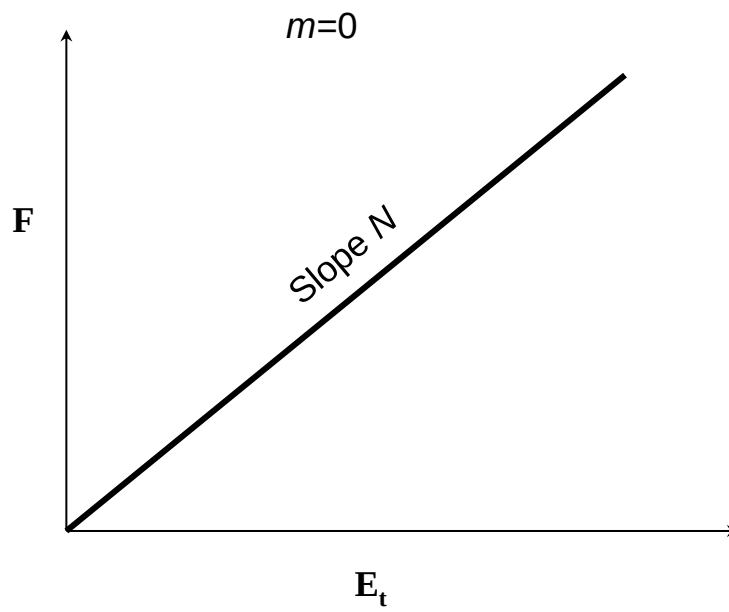
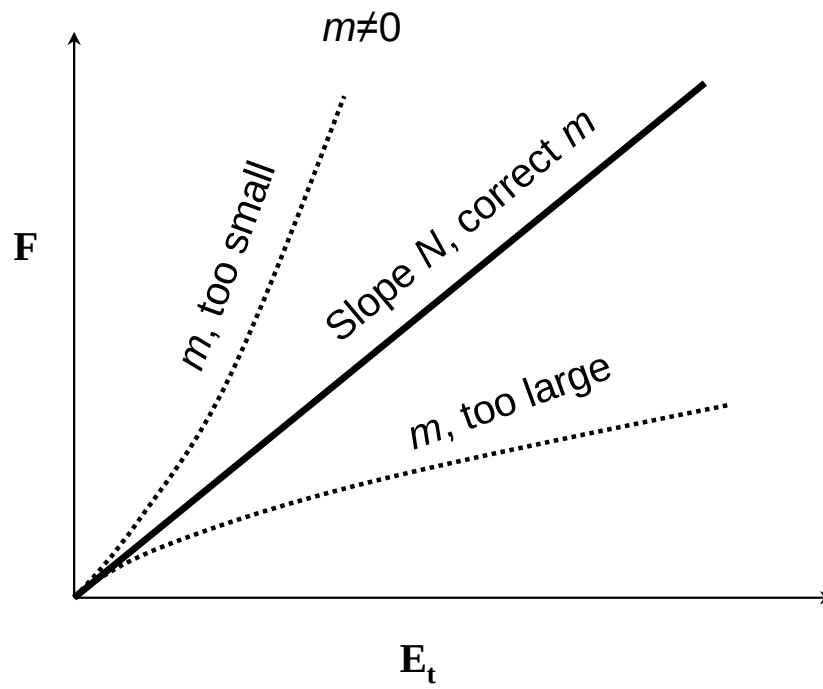
$S(p, t)$ = aquifer function

➤ **FE method (solution gas drive)**

Assume $W_e = 0$,

$$F = NE_t$$

Plot: F vs E_t through origin, N =Slope

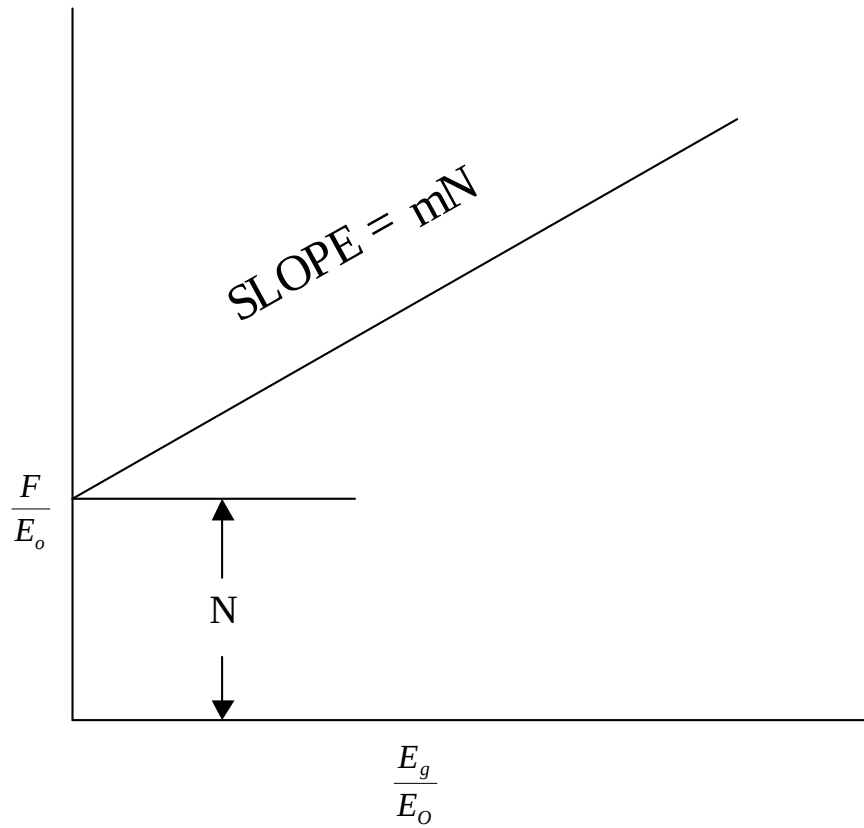


➤ **GASCAP method**

Assume $W_e=E_{fw}=0$, divided by E_o

$$\frac{F}{E_o} = N + mN \left(\frac{E_g}{E_o} \right)$$

Plot: $\frac{F}{E_o}$ vs $\frac{E_g}{E_o}$, N = Y intercept, mN = Slope

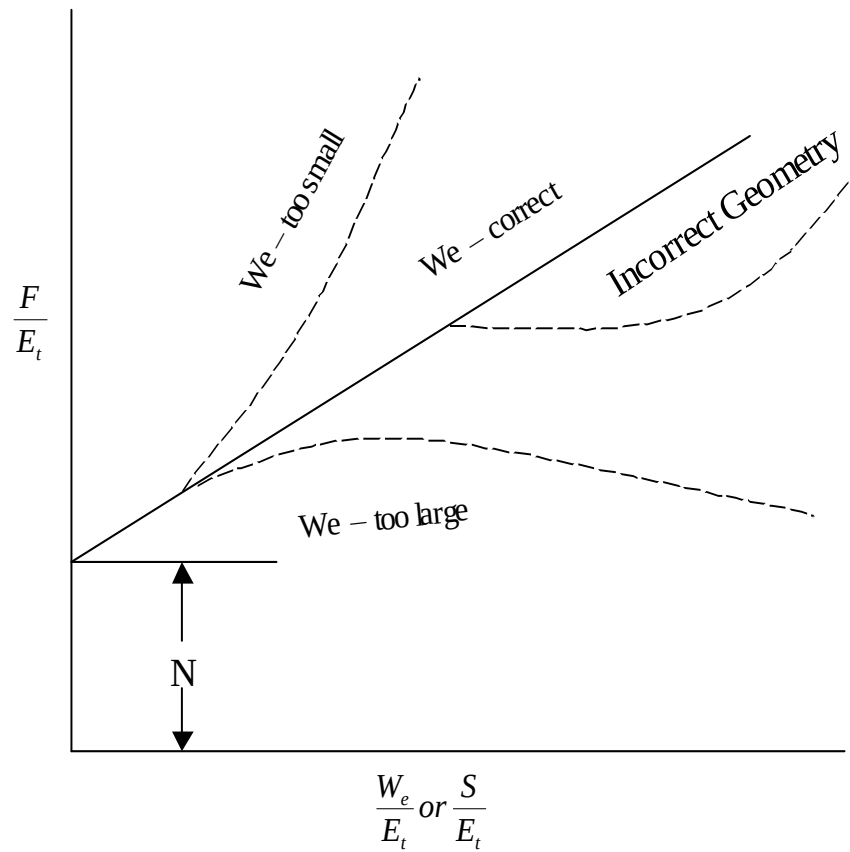


➤ **Havlena – Odeh method**

Water drive, divided by E_t

$$\frac{F}{E_t} = N + \frac{W_e}{E_t} = N + U \frac{S}{E_t}$$

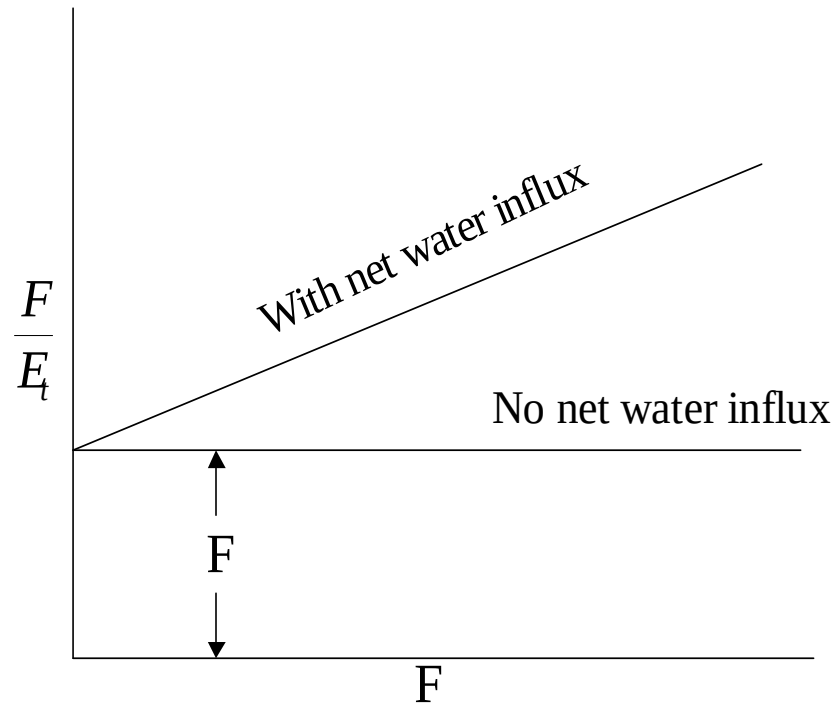
Plot: $\frac{F}{E_t}$ vs $\frac{W_e}{E_t}$ (or $\frac{S}{E_t}$), N = Y intercept, U = Slope



➤ **Campbell method**

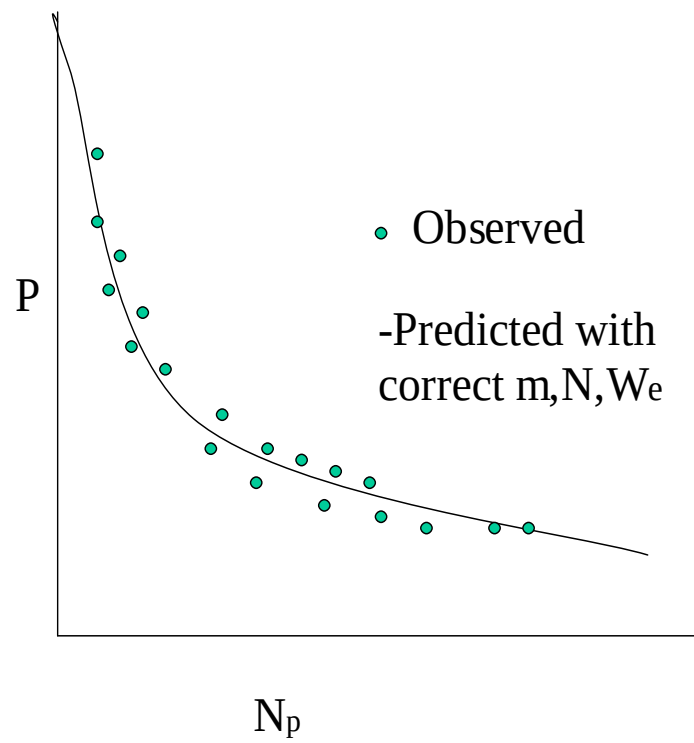
Water drive

Plot: $\frac{F}{E_t}$ vs F , $N=Y$ intercept



➤ **Pressure match methods (any reservoir)**

Plot: P_R vs N_p , Correct N , m , and W_e for best match



Production Engineering

1. Bottom Hole Pressure (BHP) in ESP Pump Wells

➤ Vertical well case

$$BHP = P_{csg} + \left\{ \frac{\rho_o}{10} (Z_{pump} - Z_{FL}) + \frac{\rho_m}{10} (Z_{pzt} - Z_{pump}) \right\}$$

Where

$\rho_m = \rho_w W_c + \rho_o (1 - W_c)$ - oil/water mixture density between the pump intake and the top depth of the pay zone.

ρ_w - water density, g/cm^3

ρ_o -- oil density, g/cm^3

Note that surface oil density is actually used in the calculation.

W_c -- water cut, *fraction*

Z 's - true vertical depths in downward direction, *m*

Z_{pump} - measured pump depth, *m*

Z_{FL} - fluid level depth, *m*

Z_{pzt} - top depth of the pay zone, *m*

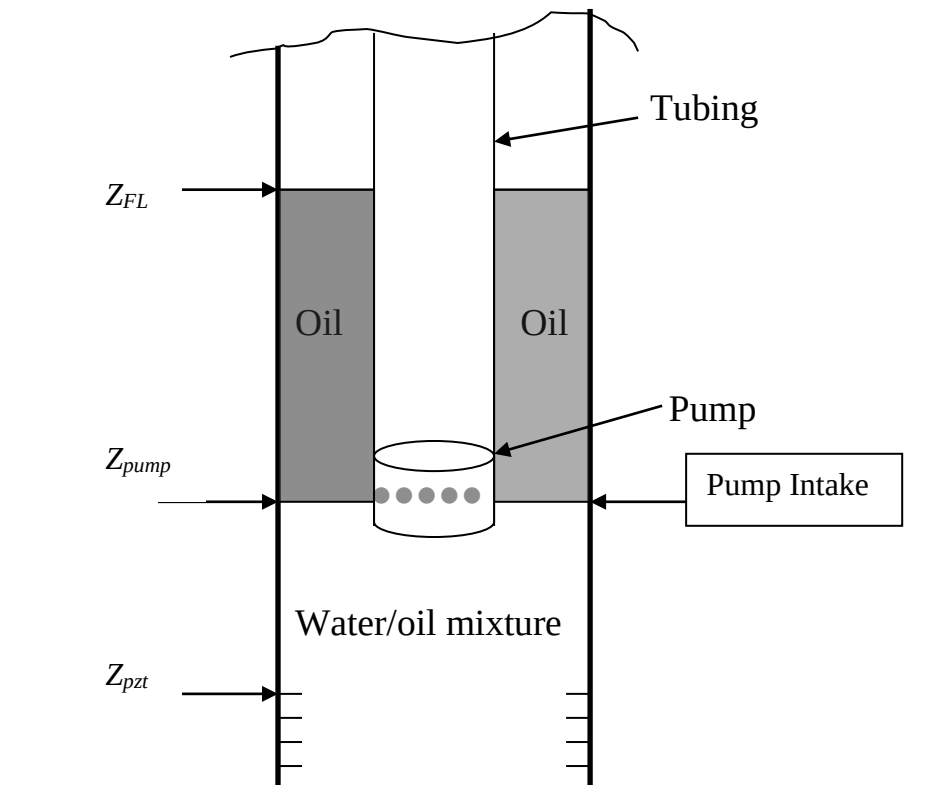
P_{csg} - casing pressure (casing/tubing annular pressure), *atm*

BHP - bottom-hole pressure, *atm or bars (kg/cm^2)*

The coefficient, 1/10, comes from the following units conversion

$$\left[\frac{g}{cm^3} \right] [m] = \left[\frac{g}{cm^3} \right] \left[\frac{kg}{1000g} \right] [m] \left[\frac{100cm}{m} \right] = \frac{1}{10} \left[\frac{kg}{cm^2} \right] = \frac{1}{10} [atm]$$

Note that it is assumed in the calculation that it is pure oil above the pump and oil/water mixture between the pump and the top of the pay zone. Further, the oil and water densities are equal to their densities at the surface conditions and the oil/water mixture density can be estimated in terms of water cut measured at surface condition.



Schematic diagram of a vertical pump well.

➤ **Deviated wells** (subtract a constant correction)

It is assumed that the deviation angle of the wellbore is constant.

$$BHP = \frac{\rho_o}{10} \left\{ (Z'_{pump} - Z'_{FL}) - \frac{|MD - TVD|}{Z'_{pzt}} (Z'_{pump} - Z'_{FL}) \right\} \\ + \frac{\rho_m}{10} \left\{ (Z'_{pzt} - Z'_{pump}) - \frac{|MD - TVD|}{Z'_{pzt}} (Z'_{pzt} - Z'_{pump}) \right\}$$

Note that $|MD - TVD|$ is termed as **deviation**, it is the difference between the measured depth and the true vertical depth of the pay zone.

Note also that all Z 's have the same meaning as in the vertical well case, but they are all measured depths along the deviated wellbore. Therefore, a prime is used to distinguish.

➤ **Units conversion**

$$1 \text{ MPa (atm)} = 14.51 \text{ psi}$$

$$1 \text{ psi} = 0.6805 \text{ atm}$$

$$1 \text{ g/cm}^3 = 62.43 \text{ lb/ft}^3$$

$$1 \text{ lb/ft}^3 = 0.01602 \text{ g/cm}^3$$

2. Oil Flow rate Based on Total Liquid and Water Cut

➤ Volumetric flowrate

$$q_o = q_L (1 - W_c), m^3/day$$

➤ Mass flowrate

$$M_o = q_L (1 - W_c) \rho_o, \text{ tones/day}$$

Where

q_L -- total liquid rate, m^3/day

ρ_o -- oil density at surface conditions, g/cm^3

W_c -- water cut, *fraction*

M_o -- oil mass rate, *tones/day*

➤ Example

A ESP well is producing $80 m^3/d$ of liquid (oil+water), surface measurement gives 16% water cut, suppose that oil density is $0.82 g/cm^3$, then the oil rates would be,

$$q_o = 80 \times (1 - 0.16) = 67.2 m^3 / d$$

$$M_o = q_o \times \rho = 67.2 \times 0.82 = 55.1 \text{ tons} / d$$

➤ Units conversion

$$1 \text{ barrel} = 0.159 m^3$$

$$1 m^3 = 6.29 \text{ barrels}$$

$$1 g/cm^3 = 62.43 \text{ lb/ft}^3$$

$$1 \text{ lb/ft}^3 = 0.01602 g/cm^3$$

3. Productivity Index PI

- To represent well productivity or deliverability

- Definition

$$PI \equiv \frac{q}{p_R - p_{wf}}$$

Where

q – flow rate, sm^3/d or stb/d

p_R – average pressure of the drainage area around the well bore, can be represented by the static fluid level, bar or psi

p_{wf} – flowing bottom-hole pressure, bar or psi

PI – productivity index, $(sm^3/d)/bar$ or $(stb/d)/psi$

- PI for steady state flow to a vertical well

$$PI = 0.05435 \times \frac{kh}{B\mu \left(\ln \frac{r_e}{r_w} + S \right)}$$

Where

kh – average kh product of the drainage area, $md.m$

r_e – drainage radius, m

r_w – wellbore radius, m

S – skin factor

B – oil FVF, m^3/sm^3

μ – oil viscosity, cp

PI – productivity index, $(sm^3/d)/bar$

- PI for pseudosteady – state flow to a vertical well

$$PI = \frac{0.05435 \times kh}{B\mu \left(\ln \frac{r_e}{r_w} - 0.75 + S \right)}$$

- PI calculation given q_L , p_{wf} and p_R

$$PI = \begin{cases} \frac{q_L}{p_R - p_b + \frac{p_b}{1.8} \left\{ 1 - 0.2 \left(\frac{p_{wf}}{p_b} \right) - 0.8 \left(\frac{p_{wf}}{p_b} \right)^2 \right\}}, & \text{if } p_{wf} \leq p_b \\ \frac{q_L}{p_R - p_{wf}}, & \text{if } p_{wf} > p_b \end{cases}$$

Where:

p_b – bubble point pressure (saturation pressure), atm

p_{wf} – flowing bottom-hole pressure, atm

p_R – reservoir pressure, atm

q_L – total liquid rate, m^3/day

PI – productivity index, $(m^3/day)/atm$

4. Inflow Performance Relationship (IPR)

➤ Vogel model in dimensionless form

$$\frac{q}{q_{\max}} = \left\{ 1 - 0.2 \left(\frac{p_{wf}}{p_{res}} \right) - 0.8 \left(\frac{p_{wf}}{p_{res}} \right)^2 \right\}$$

where q – flow rate at flowing BHP p_{wf} , less than bubble point pressure

$q_{\max}$ – maximum flow rate at $p_{wf}=0$

p_{wf} – flowing BHP

p_{res} – reservoir pressure

➤ General IPR based on Vogel's model

$$q = PI \left\{ (p_{res} - p_b) + \frac{p_b}{1.8} \left[1 - 0.2 \left(\frac{p_{wf}}{p_b} \right) - 0.8 \left(\frac{p_{wf}}{p_b} \right)^2 \right] \right\}$$

where PI – the stabilized Productivity index at zero draw down $p_{wf} \Rightarrow p_{res}$

p_b – bubble point pressure of the producing fluid

➤ General IPR based on Fetkovich's model

$$q = PI \left\{ (p_{res} - p_b) + \frac{1}{2p_b} [p_b^2 - p_{wf}^2] \right\}$$

➤ Example problem (IPR)

An oil well is producing at a stabilized rate of 20 sm^3/d at a flowing BHP $p_{wf} = 80 \text{ bars}$. The average reservoir shut-in static pressure, $p_{res} = 100 \text{ bars}$. Calculate maximum possible flow rate at $p_{wf} = 0$, and producing rate if a new pump were installed to lower flowing BHP to 50 bars . Make the calculations using Vogel's model.

1. Productivity Index (PI)

$$PI = \frac{q}{p_{res} - p_{wf}} = \frac{20}{100 - 80} = 1 (\text{sm}^3 / \text{d}) / \text{bar}$$

2. Flow rate at $p_{wf}=50\text{bars}$

$$q = PI(p_{res}-p_{wf}) = 1 \times (100-50) = 50 \text{ sm}^3/\text{d}$$

3. Vogel IPR

$$\frac{q}{q_{\max}} = 1 - 0.2 \left(\frac{80}{100} \right) - 0.8 \left(\frac{80}{100} \right)^2 = 1 - 0.16 - 0.512 = 0.328$$

$$q_{\max} = q/0.328 = 20/0.328 = 61 (\text{sm}^3/\text{d})$$

$$q|_{p_{wf}=50} = q_{\max} \left\{ 1 - 0.2 \left(\frac{50}{100} \right) - 0.8 \left(\frac{50}{100} \right)^2 \right\} = 61 \times 0.7 = 42.7 (\text{sm}^3 / \text{d})$$

5. Flow Efficiency

➤ **Definition**

$$FE = \frac{PI_{actual}}{PI_{ideal}}$$

➤ **FE for steady-state flow**

$$FE = \frac{\ln r_e / r_w}{\ln r_e / r_w + S}$$

➤ **FE for pseudosteady-state flow**

$$FE = \frac{\ln r_e / r_w}{\ln r_e / r_w - 0.75 + S}$$

6. Calculations for Enhancement Candidate Selection

➤ Enhancement

Lower the flowing bottom-hole pressure (or correspondingly dynamic fluid level) from the current higher value to 50 bars by changing a bigger size pump. Apparently, this is applicable only to producing wells with a running pump (whether it is ESP or Sucker Rod pump).

➤ Calculation of potential oil rate increment

- PI based on the current production, q_L , p_{wf} and p_{res}

- Current oil rate in *tones/day*

$q_o = q_L (1 - Wc/100) \rho_o$ where ρ_o is the oil density in g/cm^3 and Wc is the water cut in percentage

- Total liquid rate after enhancement

$q_L^{enh} = PI (p_{res} - p_{wf})$ where $p_{wf} = 50 \text{ atm}$.

- Oil rate after enhancement, in *tones per day*, assuming on change in Wc .

$q_o^{enh} = q_L^{enh} (1 - Wc/100) \rho_o$ where ρ_o is the oil density in g/cm^3 .

Note that strictly speaking, these calculations are very rough.

- Oil rate increment in *tones/day*

$\Delta q_o = q_o^{enh} - q_o$

7. Calculations for Fracturing Well Candidate Selection

A skin value of -4.7 is assumed in the calculations after fracturing.

➤ Calculation of potential oil rate increment

- kh product based the current production and skin if there is any.

$$kh = 18.4 \times \mu_o B_o \left[\ln \left(\frac{r_e}{r_w} \right) - \frac{3}{4} + S \right] * PI$$

- The new PI value after fracturing.

$$PI_{new} = \frac{kh}{18.4 \times \mu_o B_o \left[\ln \left(\frac{r_e}{r_w} \right) - \frac{3}{4} + (-4.7) \right]}$$

- Total liquid rate after fracturing

$$q_L^{frac} = PI_{new} (p_{res} - p_{wf}) \text{ where } p_{wf} = 50 \text{ atm and } p_{wf} > p_b.$$

Otherwise

$$q_L^{frac} = PI_{new} \left\{ p_{res} - p_b + \frac{p_b}{1.8} \left\{ 1 - 0.2 \left(\frac{p_{wf}}{p_b} \right) - 0.8 \left(\frac{p_{wf}}{p_b} \right)^2 \right\} \right\}, \quad p_{wf} = 50 \text{ atm}, p_{wf} < p_b$$

Note that the enhancement pressure 50atm is used in the calculations above.

- Oil rate after fracturing, in *tones per day*, assuming on change in Wc .

$$q_o^{frac} = q_L^{frac} (1 - Wc/100) \rho_o \text{ where } \rho_o \text{ is the oil density in } g/cm^3.$$

- Oil rate increment in *tones/day*

$$\Delta q_o = q_o^{frac} - q_o$$

8. ESP Pump Design

➤ Target

Identify a pump of the largest possible diameter that can be run in the well. The target efficiency should be within its recommended operating range and close to its peak efficiency

➤ Selection data

1. Mechanical Data

Casing/tubing size

Well depth /Deviation/Perforation Depth

2. Production Data

Current/potential production rate

Oil rate /water cut/GOR

Static BHP / fluid level

Producing BHP / fluid level

Bottom hole pressure

System backpressure from flowline, separator and choke

3. Fluid Data

Oil/gas/water properties

4. Power Supply

Voltage/frequency

Capacity of the service

Quality of the service

➤ Total dynamic head (TDH) for the number of stages in the pump

True vertical lift distance from the producing fluid level to the surface

Friction loss in the tubing string

Wellhead pressure

➤ Pump performance curve review for optimal producing range and peak efficiency

➤ The number of stages required, n_s

$$n_s = \frac{TDH}{L_s}$$

where TDH – the total dynamic head, m

L_s – lift per stage, m

n_s – the number of stages required

➤ Horse power required, P_{hp}

$$P_{hp} = P_{hs} \times n_s \times \rho_f$$

where

P_{hp} – motor horse power

P_{hs} – horse power per stage

n_s – number of stages

ρ_f – specific gravity of the pumping fluid

➤ Specific pump selection by computer program based on data (setting depth, casing size and motor voltage) and models (pump performance, reservoir response and fluid characteristics)

9. Fracture Efficiency

➤ Definition

$$\mathcal{E} = \frac{q_o|_{post-frac} - q_o|_{pre-frac}}{q_o|_{p_{wf}=0} - q_o|_{pre-frac}}$$

where

$q_o|_{post-frac}$ and $q_o|_{pre-frac}$ are the oil flowrate after fracture and before fracture in t/d, calculated from the total liquid rate in m3/d and water cut.

$q_o|_{p_{wf}=0}$ is called Absolute Open Flow (AOF) rate of oil at $p_{wf} = 0$, which can be calculated from the following equation,

$$q_o|_{p_{wf}=0} = PI \left(p_{res} - p_b + \frac{p_b}{1.8} \right) \left(1 - \frac{Wc}{100} \right) \rho_o = PI (p_{res} - 0.444 p_b) \left(1 - \frac{Wc}{100} \right) \rho_o$$

Remark: PI value here could be the PI before fracture, designed for a skin = -4.7 or the actual value after fracture. Correspondingly, there could be three values of $q_o|_{p_{wf}=0}$.

- **Practically** the fracture efficiency could be calculated by the actual oil rate increment divided by the designed oil rate increment

$$\mathcal{E} = \frac{q_o|_{post-frac} - q_o|_{pre-frac}}{q_o|_{skin=-4.7} - q_o|_{pre-frac}}$$

Where $q_o|_{skin=-4.7}$ is the potential oil rate based on an assumed skin factor -4.7.

10. Fracture Design Equations

➤ Rock mechanics parameters

- **Concepts**

Stress (σ) – force per unit area

Strain (ϵ) – under stress, change in length (δ), compared to the original length (l) is called strain

- **Young's modulus (E) and dynamic Young's modulus (E_d)**

$$E = \frac{\text{stress}}{\text{strain}} = \frac{\sigma}{\epsilon}, E_d = \frac{\rho}{\Delta t_s^2} \left[\frac{3\Delta t_s^2 - 4\Delta t_c^2}{\Delta t_s^2 - \Delta t_c^2} \right] \times 2.15 \times 10^8$$

- **Poisson's ratio (ν) and dynamic Poisson's ratio (ν_d)**

$$\nu = \frac{\text{lateral strain}}{\text{axial strain}}, \nu_d = \frac{1}{2} \left[\frac{\Delta t_s^2 - 2\Delta t_c^2}{\Delta t_s^2 - \Delta t_c^2} \right]$$

Typical values: sandstone (0.25), limestone (0.15), shale (0.4)

- **Shear modulus (G) and dynamic shear modulus (G_d)**

$$G = \frac{\text{shear stress}}{\text{angle of deformation}} = \frac{\sigma}{\theta \text{ in radians}}, G_d = \frac{\rho}{\Delta t_s^2} \times 2.15 \times 10^8$$

For fluid $G=0$, for solid $G=\text{finite number}$

- **Bulk modulus (K) and dynamic bulk modulus (K_d)**

$$K = \frac{\text{Force/surface area}}{\text{volume change/original volume}}, K_d = \rho \left[\frac{3\Delta t_s^2 - 4\Delta t_c^2}{3\Delta t_s^2 \Delta t_c^2} \right] \times 2.15 \times 10^8$$

K is the reciprocal of compressibility

- **Parameter relationship**

$$E = \frac{9KG}{3K + G}, \nu = \frac{E}{2G} - 1, G = \frac{3KE}{9K - E}, K = \frac{E}{3(1 - 2\nu)}$$

where Δt_s and Δt_c are shear wave and compressional wave travel time from logs, $\mu\text{s/ft}$
 ρ – rock bulk density, lb/ft^3

➤ **Fluid volume pumped (V_i)**

$$V_i = q_i \times t_p$$

Where q_i – total injection rate, m^3/hour

t_p – pumping time for a treatment, *hour*

➤ **Fracture volume (V_f)**

$$V_f = h_f \times w \times 2L = \eta \times V_i$$

Where

h_f – average, gross fracture height

w – average fracture width

L – fracture half length or penetration

η – the fluid efficiency

➤ **Volume lost into the reservoir (V_{LP})**

$$V_{LP} = 6C_L h_L L \sqrt{t_p} + 4L h_L S_p$$

where C_L – the fluid loss coefficient (typically 0.0005 to 0.05 $\text{ft}/\text{min}^{0.5}$)

h_L – fluid loss height

S_p – spurt loss (typically from 0 to 50 $\text{gal}/100\text{ft}^2$)

➤ **Fracture half length or penetration (L)**

$$L = \frac{q_i t_p}{6C_L h_L \sqrt{t_p} + 4h_L S_p + 2wh_f}$$

➤ **Dimensionless fracture conductivity (FCD)**

$$FCD = \frac{k_f w_f}{k x_f}$$

Where

x_f – fracture half-length, *m*

w_f – fracture width, *m*

k – reservoir permeability, *md*

k_f – permeability of proppant in the fracture, *md*

$k_f w_f$ – fracture conductivity, *md.m*

Note: FCD is the ratio of the ability of fracture to carry flow divided by the ability of the formation to feed the fracture.

➤ **Fracture Conductivity**

➤ **Dimensionless Pressure, p_D**

➤ **Dimensionless Time, t_D**

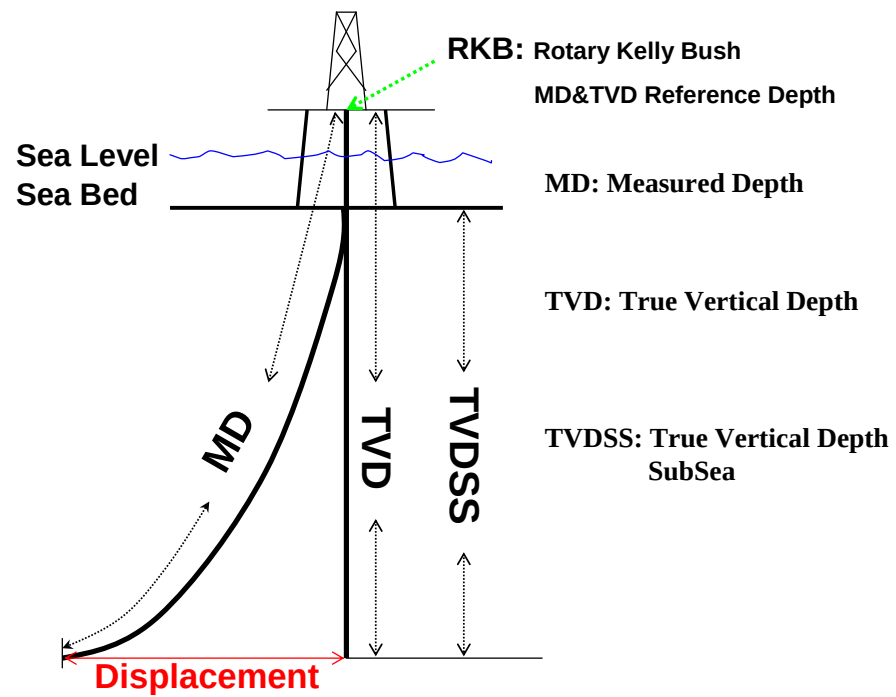
➤ **Time to Pseudo Radial Flow, t_{prf}**

➤ **Time to Pseudo Steady State Flow, t_{pss}**

➤ **Skin Factor Based on PSS p_D**

11. MD, TVD and TVDSS

MD, TVD and TVDSS



12. Single Phase Pressure Drop in Casing or Tubing

The type of the flow is characterized by the Reynolds number Re ,

$$Re = \frac{\rho v d}{\mu}$$

Where

- v - fluid velocity (m/s)
- d - inner diameter of casing or tubing (m)
- μ - dynamic viscosity (Pa.s)
- ρ - fluid density of fluid (kg/m³)

A critical value $(Re)_c = 2100$. Laminar flow: $Re \leq (Re)_c$; Turbulent flow: $Re > (Re)_c$.

For an annular

$$Re = \frac{0.8165(d_o - d_i)v\rho}{\mu}$$

Where,

- d_o - outer diameter of the annular (m)
- d_i - inner diameter of the annular (m)

The critical velocity v_c is the velocity at the critical Reynolds number. If $v \leq v_c$, the flow is of the laminar type.

➤ Pressure drop in casing or tubing

- **Critical velocity:** v_c (m/s)

$$v_c = \frac{2100\mu}{d\rho}$$

- **Critical flow rate:** q_c (m³/s)

$$q_c = \frac{525\pi d\mu}{\rho}$$

- **Pressure loss inside casing or tubing:** ΔP_i (laminar flow) (Pa)

$$\Delta P_i = \frac{32Lv\mu}{d^2} = \frac{128Lq\mu}{\pi d^4}$$

- **Pressure loss inside casing or tubing:** ΔP_i (turbulent flow) (Pa)

$$\Delta P_i = \frac{0.1L\rho^{0.8}v^{1.8}\mu^{0.2}}{d^{1.2}} = \frac{1.21L\rho^{0.8}q^{1.8}\mu^{0.2}}{\pi^{1.8}d^{4.8}}$$

where

- L - length (m)
- V - fluid velocity (m/s)
- q - flow rate (m³/s)
- d - inner diameter (m)
- μ - dynamic viscosity (Pa.s)
- ρ - density of fluid (kg/m³)

➤ **Pressure drop in annular**

- **Critical velocity**, m/min

$$v_c = \frac{6.08\mu}{(d_o - d_i)\rho}$$

- **Pressure drop** ΔP_i (laminar flow) (bar)

$$\Delta P_i = \frac{Lv\mu}{80645(d_o - d_i)^2} = \frac{Lq\mu}{40863(d_o + d_i)(d_o - d_i)^3}$$

- **Pressure drop** ΔP_i (turbulent flow) (bar)

$$\Delta P_i = \frac{L\rho^{0.8}v^{1.8}\mu^{0.2}}{240415(d_o - d_i)^{1.2}} = \frac{L\rho^{0.8}q^{1.8}\mu^{0.2}}{70696(d_o + d_i)^{1.8}(d_o - d_i)^3}$$

where

L - length (m)

v - fluid velocity (m/min)

q - flow rate ($liter/min$)

d_o - outer diameter (m)

d_i - inner diameter (m)

μ - fluid dynamic viscosity (cp)

ρ - fluid density (kg/dm^3)

13. Flow through Chokes

Control production rate from a well. It should be selected so that the fluctuations in the line pressure downstream of the choke have no effect on the well flowrate.

➤ **Gilbert equation (water/oil, producing gas)**

$$p_t = \frac{435GOR^{0.546}q_t}{S_c^{1.89}}$$

where

p_t – tubing pressure, *psi*

GOR – gas/liquid ratio, $10^3 scf/bbl$

q_t – gross liquid rate, *b/d*

S_c – choke size in $1/64$ in

➤ **Example**

It is desired to produce a well at 100b/d oil, the producing GOR is 4×10^3 *scf/bbl*. At this rate, the tubing pressure is 1800psi, estimate the choke size.

$$S_c^{1.89} = \frac{435GOR^{0.546}q_t}{p_t} = \frac{435 \times 4^{0.546} \times 100}{1800} = 0.87$$

$S_c = 7.4$ (1/64 in), a 8/64in choke can be used

➤ **Poettmann and Beck's charts**

For flow rate estimation given choke size, GOR and p_t

From GOR to Tubing pressure to choke size, then find q_t

For choke size estimate given others

From q_t to p_t to GOR horizontally, then choke size

14. Multiphase Pressure Drop in Casing or Tubing

Multiphase flow in vertical pipes may be categorized into four different flow regimes, which are bubble flow, slug flow, transition flow and mist flow. Complete sets of pressure traverses for specific flow conditions, tubing/casing sizes and fluid properties have been published and can be used for hand calculations.

Here Orkiszewski's correlations for bubble flow and slug flow are given only as reference. Due to this, English units will be applied to the equations.

➤ General pressure drop expression

$$\Delta p = \frac{1}{144} \left\{ \frac{\rho_{avg} + \tau_f}{1 - \frac{w_t q_g}{144 g_c A^2 p_{avg}}} \right\} \Delta Z$$

where

- ρ_{avg} – average fluid density, lbm/ft^3
- τ_f – friction loss gradient, lbf/ft^2-ft
- w_t – total mass flowrate, lbm/sec
- q_g – gas or light phase volumetric flowrate, ft^3/sec
- A – cross-sectional area of the pipe, ft^2
- p_{avg} – average pressure in ΔZ , psi
- g_c – gravitational constant, $32.2 ft-lbm/lbf-sec^2$
- ΔZ – depth interval for the segment, ft
- Δp – pressure drop in ΔZ , psi

Remark: the key to Δp calculation is to evaluate ρ_{avg} and τ_f for different flow regimes.

➤ Orkiszewski's correlation

➤ Dimensionless variables for flow regime determination

Dimensionless gas velocity

$$v_{gD} = \frac{q_g [\rho_L / (g\sigma)]^{0.25}}{A}$$

at the bubble slug boundary

$$L_B = 1.071 - \frac{0.2218 v_t^2}{d_H}$$

but $L_B \geq 0.13$

at the slug-transition boundary

$$L_S = 50 + \frac{36 v_{gD} q_L}{q_g}$$

and at the transition-mist boundary

$$L_M = 75 + 84 \left(\frac{v_{gD} q_L}{q_g} \right)^{0.75}$$

where

- v_{gD} - dimensionless gas velocity
- v_t - total fluid velocity (q_t/A), ft/s
- ρ_L - liquid density, $lbm/cu ft$

σ - liquid surface tension, lbm/sec^2
 L - flow regime boundary, dimensionless
 d_H - hydraulic pump diameter, ft
 d_g - gas flow rate, $cu\ ft/sec$
 g - acceleration of gravity, ft/sec^2
 A - flow area of pipe, $sq\ ft$

➤ Bubble flow

The average flowing density in bubble flow is calculated from the following equation, which volumetrically weights the gas and liquid densities.

$$\bar{\rho}_g = \rho_g f_g + (1 - f_g) \rho_L$$

The following gas fraction f_g in bubble flow is given by

$$f_g = \frac{1}{2} \left[1 + \frac{q_t}{v_s A} - \sqrt{\left(\frac{1 + q_t}{v_s A} \right)^2 - \frac{4q_g}{v_s A}} \right]$$

where the slip velocity, v_s is the difference between the average gas and liquid velocity. Griffith suggests the use of an approximate value of $v_s = 0.8\ ft/sec$ for bubble flow

The friction loss gradient for bubble flow is based on single-phase liquid flow.

$$\tau_f = \frac{f \rho_L v_L^2}{2g_c d_H \cos \theta}$$

where

$$v_L = \frac{q_L}{A(1 - f_g)}$$

$$N_{Re} = \frac{1488 \rho_L d_H v_L}{\mu_L}$$

Where

d_H - the hydraulic pipe diameter, ft

μ_L - Liquid viscosity

➤ Slug flow

Average density term for *slug flow* is expressed as

$$\bar{\rho} = \frac{w_t + \rho_L v_b A}{q_t + v_b A} + \delta \sigma_L$$

v_b - bubble rise velocity

$$v_b = C_1 C_2 \sqrt{g d_H}$$

$$N_{Re_b} = \frac{1488 \rho_L d_H v_b}{\mu_L}$$

$$N_{Re} = \frac{1488 \rho_L d_H v_t}{\mu_L}$$

$$v_b = [0.546 + 8.74(10^{-6}) N_{Re}] \sqrt{g d_H}$$

$$v_b = 0.5 v_{bi} + 0.5 \left(v_{bi}^2 + \frac{13.59 \mu_L}{\rho_L \sqrt{d_H}} \right)^{0.5}$$

$$v_{bi} = \left[0.251 + 8.74(10^{-6})N_{Re} \right] \sqrt{gd_H}$$

$$v_b = \left[0.35 + 8.74(10^{-6})N_{Re} \right] \sqrt{gd_H}$$

$$\tau_f = \frac{f\rho_L v_b^2}{2g_c d_H \cos\theta} \left(\frac{q_L + v_b A}{q_t + v_b A} + \delta \right)$$

➤ **Continuous oil phase**

When $v_t < 10$

$$\delta = \frac{0.0127}{d_H^{1.415}} \log(\mu_L + 1) - 0.284 + 0.167 \log v_t + 0.113 \log d_H$$

When $v_t > 10$,

$$\delta = \frac{0.0274}{d_H^{1.371}} \log(\mu_L + 1) - 0.161 + 0.569 \log d_H$$

$$- \log v_t \left[\frac{0.01}{d_H^{1.571}} \log(\mu_L + 1) + 0.397 + 0.63 \log d_H \right]$$

➤ **Continuous water phase**

When $v_t < 10$,

$$\delta = \frac{0.013}{d_H^{1.8}} \log \mu_L - 0.681 + 0.232 \log v_t - 0.428 \log d_H$$

When $v_t > 10$,

$$\delta = \frac{0.045}{d_H^{0.799}} \log \mu_L - 0.709 + 0.162 \log v_t - 0.888 \log d_H$$

15. Oil PVT Properties

➤ Oil specific gravity and API gravity

$$\gamma_o = \frac{141.5}{131.5 + {}^\circ API}, \quad {}^\circ API = \frac{141.5}{\gamma_o} - 131.5$$

where

γ_o – oil specific gravity

${}^\circ API$ – oil API gravity

➤ Bubble pressure or saturation pressure

$$P_b = 18.2 \left\{ \left(\frac{R_s}{r_g} \right)^{0.83} \times 10^{0.00091T_R - 0.0125{}^\circ API} - 1.4 \right\}$$

where

R_s – produced gas/oil ratio, *scf/bbl*

γ_g – gas specific gravity

T_R – reservoir temperature, ${}^\circ F$

➤ Dead oil viscosity

$$\mu_{do} = \left\{ 0.32 + \frac{1.8 \times 10^7}{{}^\circ API^{4.53}} \right\} \left(\frac{360}{T_R + 200} \right)^D$$

where

$$D = 10^{0.43 + \frac{8.33}{{}^\circ API}}$$

μ_{do} – dead oil (gas free) viscosity, *cp*

and T_R – reservoir temperature, ${}^\circ F$

➤ Oil viscosity above bubble point pressure

$$\mu_o = \mu_{ob} + e^A \left(\frac{P_R - P_b}{1000} \right), \quad A = -2.68 + 0.98 \ln(\mu_{ob}) + 0.91 (\ln(\mu_{ob}))^2$$

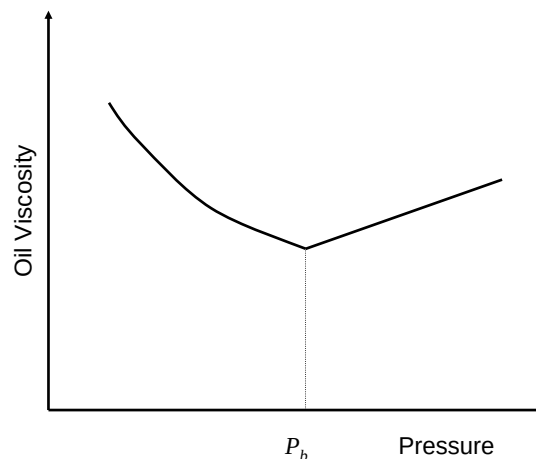
where

μ_{ob} –oil viscosity at bubble point pressure P_b , *cp*

μ_o –oil (undersaturated) viscosity, *cp*

P_R – reservoir pressure, *psi*

P_b – bubble point pressure, *psi*



➤ **Oil formation volume factor (Standing)**

At or below P_b : $B_{ob} = 0.972 + \frac{1.47}{10^4} \left[R_s \sqrt{\frac{\gamma_g}{\gamma_o}} + 1.25 T_R \right]^{1.175}$

$$B_o = B_{ob} \text{Exp}\{-C_o(P_b - p_R)\}$$

where

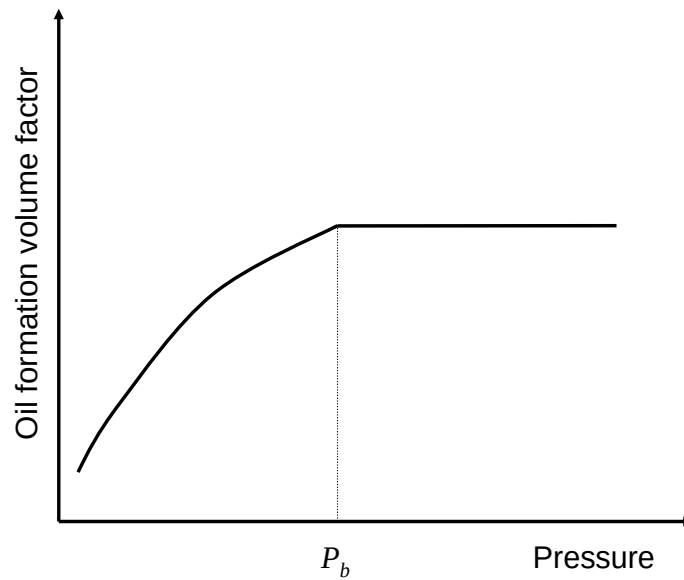
B_{ob} – oil FVF at or below bubble point pressure, *RB/STB*

B_o – oil FVF above P_b , *RB/STB*

C_o – oil compressibility, psi^{-1}

γ_g - gas specific gravity

T_R – reservoir temperature, $^{\circ}\text{F}$



16. Gas Properties

➤ Real gas equation of state (EOS)

$$pV = znRT = z \frac{m}{M} RT$$

where

p – absolute pressure (measured from vacuum), *bar*

T – absolute temperature (measured from absolute zero), $^{\circ}K = ^{\circ}C + 273$

V – volume occupied at p and T by $n = m/M$ moles of gas, m^3

m – mass of gas considered, *g*

M – molecular weight of gas

R – gas constant, $8.315 \times 10^{-5} (m^3 \cdot bar) / mol \cdot g$

z – compressibility factor

➤ Gas FVF

$$B_g = 0.00350958 \frac{zT}{p}$$

T – temperature, $^{\circ}K$

p – pressure, *bar*

B_g – gas formation volume factor, *res m^3 / std m^3*

➤ Gas viscosity

$$\mu_g = \frac{Y1 \times \exp(Y5)}{10^4}$$

where

$$Y1 = \frac{(9.4 + 0.02M)(1.8T + 492)^{1.5}}{209 + 19M + 1.8T + 492}$$

$$Y5 = Y2(Y4)^{Y3}$$

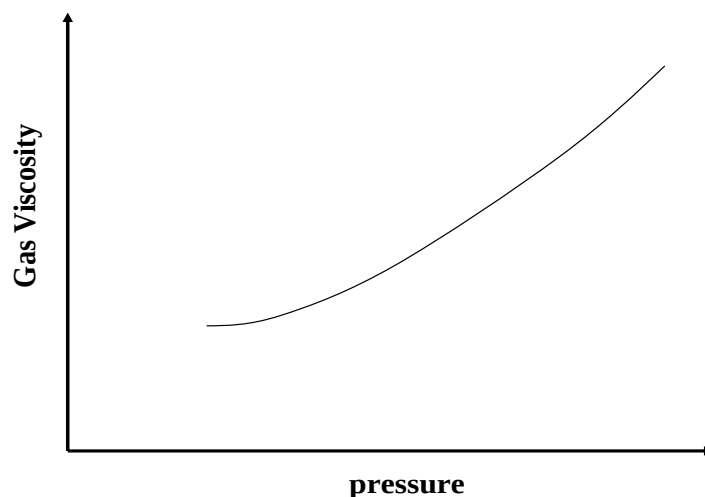
$$Y4 = 0.007532 \frac{M}{B_g}$$

$$Y3 = 2.4 - 0.2Y2$$

$$Y2 = 3.5 + 0.01M + \frac{986}{1.8T + 492}$$

$$M = 28.9 \gamma_g \text{ (Lee et al.)}$$

μ_g – gas viscosity, *cp*



➤ **z factor**

- **Definition**

$$z = \frac{V_{actual}}{V_{ideal}}$$

where V_{actual} – actual volume of gas at a specified temperature (T) and pressure (p)
 V_{ideal} – ideal volume of gas at the same T and p

- **Correlation**

$$z = 1 + \left(A_1 + \frac{A_2}{T_{pr}} + \frac{A_3}{T_{pr}^3} + \frac{A_4}{T_{pr}^4} + \frac{A_5}{T_{pr}^5} \right) \rho_{pr} + \left(A_6 + \frac{A_7}{T_{pr}} + \frac{A_8}{T_{pr}^2} \right) \rho_{pr}^2 - A_9 \left(\frac{A_7}{T_{pr}} + \frac{A_8}{T_{pr}^2} \right) \rho_{pr}^5 + A_{10} \left(1 + A_{11} \rho_{pr}^2 \right) \left(\frac{\rho_{pr}^2}{T_{pr}^3} \right) \exp(-A_{11} \rho_{pr}^2)$$

Where

$$\rho_{pr} = \frac{0.27 p_r}{(z T_r)}, \quad T_{pr} = \frac{T}{T_{pc}}, \quad P_{pr} = \frac{p}{P_{pc}}$$

P_{pc} and T_{pc} – pseudo critical pressure and temperature which can be calculated by

$$P_{pc} = 756.8 - 131.07 \gamma_g - 3.6 \gamma_g^2$$

$$T_{pc} = 169.2 + 349.5 \gamma_g - 74.0 \gamma_g^2$$

with γ_g – gas specific gravity ranging from 0.57 to 1.68.

and constants A_1 through A_{11} are

$$A_1 = 0.3265; A_2 = -1.700; A_3 = -0.5339; A_4 = -0.01569; A_5 = -0.01565 \\ A_6 = -0.5475; A_7 = -0.7361; A_8 = 0.1844; A_9 = 0.1056; A_{10} = 0.6134; \\ A_{11} = 0.7210$$

17. Decline Curve Equations

➤ General mathematical expression

$$D = -\frac{dq/dt}{q} = Kq^n$$

Where,

q - production rate, cubic meters per day, month or year

t - time, day, month

K - constant

n - exponent

Note: The decline rate in this equation can be constant or variable with the time yielding three basic types of production decline.

➤ Exponential or constant decline

$$D = -\frac{dq/dt}{q} = K = -\frac{\ln\left(\frac{q_t}{q_i}\right)}{t}$$

where

$n = 0$, K – constant

q_i – initial production rate

q_t – production rate at time t

The rate- time and rate – cumulative relationships are given respectively by

$$q_t = q_i e^{-Dt}, \quad Q_t = \frac{q_i - q_t}{D}$$

A familiar rate constant for exponential decline is as follows:

$$D' = \frac{\Delta q}{q_i}$$

Where Δq is the rate change in the first year. In this case, the relationship between D and D' is given below.

$$D = -\ln\left(1 - \frac{\Delta q}{q_i}\right) = -\ln(1 - D')$$

➤ Hyperbolic decline

$$D = -\frac{dq/dt}{q} = Kq^n \quad (0 < n < 1)$$

Note that this is the same equation as the general decline rate equation except for the constraint on n . For initial condition

$$K = \frac{D_i}{q_i^n}$$

The rate – time and rate – cumulative relationships are given by:

$$qt = q_i (1 + nD_i t)^{-\frac{1}{n}}, \quad Q_t = \frac{q_i^n (q_i^{1-n} - q_t^{1-n})}{(1-n)D_i}$$

Where D_i = initial decline rate

➤ **Harmonic decline**

$$D = -\frac{dq/dt}{q} = Kq$$

Where $n = 1$ and for initial condition

$$K = \frac{D_i}{q_i}$$

The rate – time and rate – cumulative relationships are given by:

$$q_t = \frac{q_i}{(1 + D_i t)}, \quad Q_t = \frac{q_i}{D_i} \ln \frac{q_i}{q_t}$$

Note also that both harmonic and exponential declines are special cases of the hyperbolic decline.

Formation Evaluation

1. Shale Volume by GR and SP

➤ **Definition: shale content, V_{sh}**

$$V_{sh} = \frac{\text{Shale Volume}}{\text{Rock Volume}} \times 100\%$$

➤ **GR – natural gamma ray log**

➤ Relative GR for shale content

$$V_{sh} = \Delta GR = \frac{GR - GR_{sand}}{GR_{shale} - GR_{sand}} \times 100\%$$

Where

GR – GR of reservoir

GR_{sand} – GR of thick clean sand without hydrocarbon, minimum

GR_{shale} – GR of pure shale, maximum

➤ Non-linear correction

$$V_{sh} = \frac{2^{\Delta GR \times c} - 1}{2^c - 1} \times 100\%$$

Where coefficient c depends on formation age,

Jurassic formation $c=3.7$, older formation $c=2$

➤ **SP – spontaneous potential**

$$V_{sh} = \Delta SP = \left(1 - \frac{SP}{SSP} \right) \times 100\%$$

where

SP – SP of the reservoir

SSP – Static SP in thick/clean sand without hydrocarbon content.

2. Porosity from Porosity Logs

➤ Porosity of clean reservoirs ($V_{sh}=0$)

- Neutron Porosity

$$\varphi_N = \frac{\Phi_N - \Phi_{Nma}}{\Phi_{Nmf} - \Phi_{Nma}}$$

Φ_N – neutron of reservoir in limestone porosity units, percentage

Φ_{Nma} – matrix neutron (sandstone: -4~-6%; limestone:0; dolomite: 2%)

Φ_{Nf} – neutron of mud filtration, approximately 100%

- Density Porosity

$$\varphi_D = \frac{\rho_b - \rho_{ma}}{\rho_{mf} - \rho_{ma}}$$

ρ_b – density of reservoir, g/cm³

ρ_{ma} – matrix density (sandstone: 2.65; limestone:2.71; dolomite: 2.87)

ρ_{mf} – density of mud filtration, approximately 1g/cm³

- Acoustic Porosity

$$\varphi_S = \frac{\Delta t - \Delta t_{ma}}{\Delta t_f - \Delta t_{Nma}} \frac{1}{C_p}$$

Δt – time interval of the reservoir formation, $\mu s/m$

Δt_{ma} – transit time of matrix (sandstone: 182.1; limestone:155.8; dolomite: 142.7)

Δt_{mf} – transit time of mud filtration, approximately, 680 $\mu s/m$.

C_p – compaction factor, can be estimated from $\Delta t_{sh}/30.48$

Where Δt_{sh} is the transit time in adjacent shale.

➤ Porosity of shaly reservoirs, shale content correction

- Neutron Porosity

$$\varphi_N = \frac{\Phi_N - \Phi_{Nma}}{\Phi_{Nmf} - \Phi_{Nma}} - V_{sh} \frac{\Phi_{Nsh} - \Phi_{Nma}}{\Phi_{Nmf} - \Phi_{Nma}}$$

- Density Porosity

$$\varphi_D = \frac{\rho_b - \rho_{ma}}{\rho_{mf} - \rho_{ma}} - V_{sh} \frac{\rho_{sh} - \rho_{ma}}{\rho_{mf} - \rho_{ma}}$$

- Acoustic Porosity

$$\varphi_S = \frac{\Delta t - \Delta t_{ma}}{\Delta t_f - \Delta t_{Nma}} \frac{1}{C_p} - V_{sh} \frac{\Delta t_{sh} - \Delta t_{ma}}{\Delta t_{mf} - \Delta t_{ma}}$$

3. Archie's Equations

- Link resistivity log value to water saturation and porosity log value, experimental results, for saturation calculation from formation resistivity
- Formation factor

$$F \equiv \frac{R_o}{R_w} \approx \frac{a}{\phi^m}$$

where

a - lithology coefficient (1 in general is a good estimate) and

m - porosity cementation index (≈ 2 , in general)

R_w - formation water resistivity, Ωm , can be calculated from SP

R_o - resistivity 100% water-saturated formation with water resistivity R_w , Ωm

ϕ - porosity determined by porosity logs (Acoustic/Neutron/Density), fraction

F - formation factor (constant for a given sample)

Useful models are listed

1) Commonly used sandstone equations (Humble) $F = \frac{0.625}{\phi^{2.15}}$, $F = \frac{0.81}{\phi^2}$

2) Compact rocks, low porosity, non-fractured carbonate

$$F = \frac{1}{\phi^{2.2 \sim 2.5}}, \quad F = \frac{1}{\phi^{1.87 + 0.019/\phi}}$$

3) Chevron equation for sandstone $F = \frac{1.13}{\phi^{1.73}}$

- Formation resistivity index, I

$$I \equiv \frac{R_t}{R_o} = \frac{1}{S_w^n}$$

or combined with F equations for a S_w equation

$$S_w = \sqrt[n]{\frac{R_o}{R_t}} = \sqrt[n]{\frac{FR_w}{R_t}} = \sqrt[n]{\frac{aR_w}{\phi^m R_t}}$$

Where

R_t - the true formation resistivity, Ωm

n - saturation coefficient, ≈ 2 in general.

S_w - water saturation of the formation, fraction

Note that the Archie equations are satisfied for clean formations (sandstone with very little shale content). For shaly-sand, shale correction is needed.

- Example

A clean consolidated sandstone formation has $R_t = 10\Omega m$, if the porosity and water resistivity are 15% and $0.05\Omega m$ respectively, Calculate R_o and S_w and determine whether the formation is oil-bearing?

Solution: use $F = \frac{1.13}{\phi^{1.73}}$ for example, $F = 1.13/0.15^{1.73} = 30$

$R_o = FR_w = 30 \times 0.05 = 1.5\Omega m \ll R_t$ the formation is possibly oil-bearing.

$S_w = (R_o/R_t)^{0.5} = 0.387$ (38.7%)

Oil saturation = $1 - S_w = 61.3\%$

➤ **Correction for shale content (V_{sh})**

- Simandoux equation

$$S_w = \frac{0.4R_w}{\varphi_e^2} \left\{ -\frac{v_{sh}}{R_{sh}} + \sqrt{\frac{v_{sh}^2}{R_{sh}^2} + \frac{5\varphi_e^2}{R_t R_w}} \right\}$$

where

v_{sh} – shale content determined by shale indicator curves (SP , GR , etc),
fraction

R_{sh} – shale resistivity taken from the neighboring shale, Ωm

φ_e – effective porosity that excludes the shale effect

- Fertle-Hammack equation

$$S_w = \sqrt{\frac{FR_w}{R_t}} - \frac{v_{sh}R_w}{0.4R_{sh}\varphi_e}$$

where F reflects effective porosity too. Other parameters are the same as above.

➤ **Example**

A shaly sandstone reservoir has the following log data and parameters:

$$R_t = 5\Omega m, R_w = 0.05\Omega m, V_{sh} = 20\%, R_{sh} = 1\Omega m \text{ and } \varphi_e = 18\%$$

1. Use Archie equation for F and water saturation S_w

$$F = 0.81/\varphi_e^2 = 0.81/0.18^2 = 25$$

$$S_w = (FR_w/R_t)^{0.5} = 25 \times 0.05/5 = 0.5 \text{ (50\%)}$$

2. Use Fertle-Mammack equation to account for the shale effect

$$S_w = (FR_w/R_t)^{0.5} - v_{sh}R_w/(0.4R_{sh}\varphi_e) = 0.5 - 0.2 \times 0.05/(0.4 \times 0.18 \times 1) \\ = 0.5 - 0.14 = 0.36 \text{ (36\%)}$$

3. Use Simandoux equation for shale correction

$$S_w = \frac{0.4(0.05)}{0.18^2} \left\{ -\frac{0.2}{1} + \sqrt{\frac{0.2^2}{1^2} + \frac{5 \times 0.18^2}{0.05 \times 5}} \right\} = 0.388 \text{ (38.8\%)}$$